

Machine-Learning–Augmented Linear Multistep Methods: Matching High-Order Runge–Kutta Accuracy and Stability at Linear Multistep Cost

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Abstract. The quality of a time-integration scheme is set by three properties: cost, stability, and accuracy. Cost splits into memory and run time. Accuracy is set by the order and by the local truncation error (LTE). Linear multistep methods (LMMs) are the workhorse of pseudo-spectral turbulence codes. The reason is structural. An LMM that treats the nonlinearity explicitly reuses its stored history, so each step costs one new nonlinear evaluation. In an implicit–explicit (IMEX) pairing, the stiff linear operator is diagonal in Fourier space, and the implicit solve is an elementwise division. Runge–Kutta (RK) methods offer higher order and larger stability regions. But an RK method rebuilds its stages at every step: an s -stage method pays s nonlinear evaluations, four for classical RK4, and for orders beyond four more stages than the order are required. This paper constructs a *family of ML-augmented LMMs* with high-order RK accuracy and stability, at LMM cost. The user chooses the LMM, its order p , and the RK- q reference. We showcase the methodology on AB2–CN2 versus RK4 in two-dimensional quasi-geostrophic turbulence. The correction that moves an LMM onto an RK trajectory is, by construction, the scheme’s truncation error. We therefore derive the LTE of AB2–CN2 in closed form to $\mathcal{O}(\Delta T^6)$, and compare it term by term with those of AB4–CN2 and RK4. Each truncation operator splits in two. The analytical part, powers of the Fourier-diagonal linear operator, is free at inference. The expensive part is built from time derivatives of the advective Jacobian. A compact $\mathcal{O}(10^3)$ -parameter physics-initialized network estimates only the expensive part, from a depth- S stencil of stored states. The estimator differentiates this history by finite differences in time, so its error can be analyzed exactly. At every order the error is driven by a single field. A Wiener (least-squares filter) analysis then shows the optimal learned correction depends on step size and flow regime only through $\Delta T \sigma(\kappa)$, a per-shell decorrelation rate measurable from the model’s own inputs. Conditioning the stencils on this rate lets one network generalize across arbitrary physics (Re , β , μ) and step sizes. Stability is built in rather than hoped for. The correction matches its target only up to a finite order, so the ML-augmented scheme keeps its own higher-order dissipation. Matching RK4 adds RK4’s provable $y^6/144$ margin on the advection axis. The analytic closure delivers RK-grade accuracy and stability at LMM cost. The learned estimator is what makes it cheap.

Keywords. time-integration error, modified equation, closure modeling, IMEX schemes, Wiener filtering, von Neumann stability, pseudo-spectral methods, quasi-geostrophic turbulence, scientific machine learning